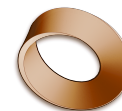


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Design



1.1 – Introduction to MakeCode

Unit 1: Robotics & Control

1. **Open.** `makecode.microbit.org` → **New Project** → name it name-badge.
2. **First letter.** Drag `show leds` inside `on start`. Click each LED you want lit.
3. **Download.** USB in → purple **Download**. If a file lands, drag it onto the **MICROBIT** drive.
Stop here until your board shows the letter.
4. **Your name.** One `show leds` per letter, with `pause (ms) 100` after each. Finish with a *blank* `show leds` and `pause (ms) 500` — that is the gap before it starts again.
5. **Loop it.** Wrap everything in `repeat 4 times`. A repeat block is called a **loop** — it runs the same blocks over and over.
6. **Add an input.** Put it all inside `on button A pressed` instead of `on start`.
button = **input** loop = **process** LEDs = **output**
7. **Publish.** **Share** → **Publish project** → copy the URL. Publishing is how you hand work in — you will need it for the homework.

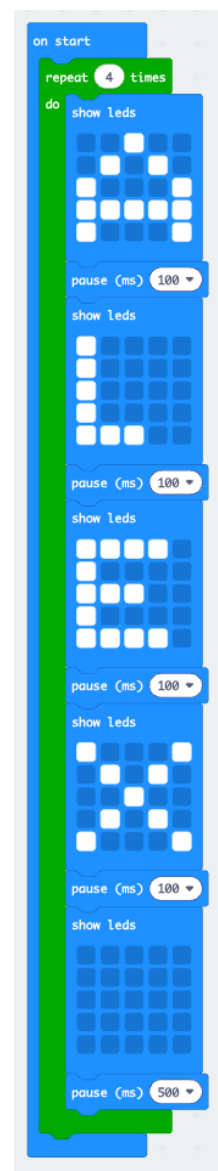
Stuck?

- **Board blank?** Cable in at both ends. Redo step 3.
- **Letters flash past?** Bigger pause. $1000 = 1$ second.
- **Letter won't fit?** The grid is 5×5 . E, L, T, I are easiest.
- **Old program gone?** The board holds one at a time. Normal.
- **Lost your link?** **Share** → **Publish** again.

Extension

Make `on button B pressed` do something different. Shrink the pauses until your name is unreadable — where is the limit? Try `show string`. What is it doing for you?

The finished program



A four-letter name,
cycled four times.

HOMEWORK

Make a **new** program that spells a word or a short message of your own.

1. Build it in MakeCode, the same way as above.
2. **Share** → **Publish project** → copy the URL.
3. **Submit the URL to Google Classroom.**